Branch Locus of Collatz Dynamics on Finite Tori: Visualizing the 100-Billion Computation

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Abstract

We present the results of an exhaustive branch locus computation for the Collatz map on the family of finite tori $\mathbb{T}_k^2 = (\mathbb{Z}/k\mathbb{Z})^2$, carried out for all $n \leq 10^{11}$ across 27 hierarchically nested grid levels ($k \in \{1, 2, 3, \ldots, 19683\}$). The computation consumed 61.5 hours on a 24-core Intel Core Ultra 9 275HX, accumulating 25.1×10^{12} Collatz steps. Eight figures document the principal findings: cell saturation, torus heatmaps, Diophantine signatures, shift-of-finite-type (SFT) forbidden transitions, foliation enrichment collapse, cell-level statistics, convergence over N, and transition/shadow analysis. A comparison with the earlier 10^{10} run reveals new structure invisible at lower N: a "Diophantine ghost island" of 251 cells at perpendicular distance 370 from the main branch locus, created by the best rational approximant $460/729 \approx \log_3 2$, together with a sharpened gap in the shadow offset distribution and a new downward-sloping feature in the foliation-position shadow profile.

1 Overview of the computation

The *branch locus* of the Collatz map at resolution k is the set of cells $(r_2, r_3) \in (\mathbb{Z}/k\mathbb{Z})^2$ that are visited by trajectories with *both* even and odd steps. The complementary cells—*pure even* (only even steps land there) and *pure odd* (only odd steps)—carry deterministic parity and are therefore confined to a single leaf of the Anosov-like foliation.

For each starting value $n = 1, 2, \ldots, 10^{11} - 1$, we iterate the Collatz map $T(n) = n/2$ (if even) or $T(n) = 3n + 1$ (if odd), and at every step t we record the residue class $(r_2, r_3) = (\nu_2(t) \bmod k,\ \nu_3(t) \bmod k)$ for each of the 27 grid levels $k = 3^a \cdot 2^b$ with $0 \leq a, b$ and $k \leq 19{,}683$. We also record transition types (even→even, even→odd, etc.) and foliation/shadow geometry.

Key parameters:

Total trajectories	99,999,999,999
Total Collatz steps	25,125,922,364,266
Mean trajectory length	251.26 steps
Global p_{odd}	0.332473361
Number of k-levels	27
Saturated branch cell count	21,632 (from $k = 216$ onward)
Runtime	61.5 hours (24-thread OpenMP)

2 Figure 1: Cell saturation and scale overview

The saturation of branch cells at exactly 21,632 is the central empirical fact. It means the torus $\mathbb{T}_k^2$ for $k \geq 216$ contains a *fixed* set of 21,632 cells that carry all the dynamical complexity. The

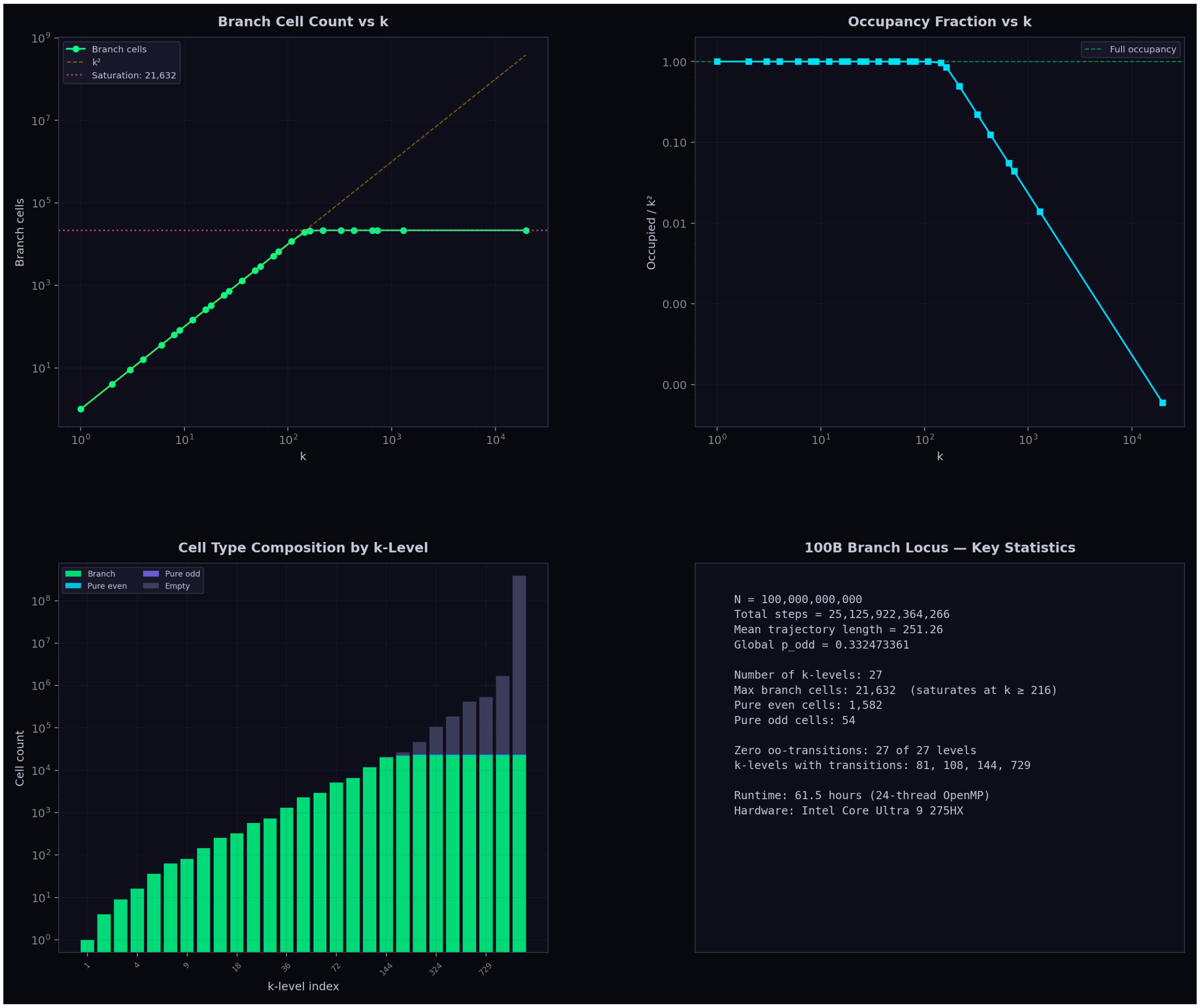

Figure 1: Cell saturation and scale overview. **(a)** Branch cell count vs. k on log–log axes. At small k every cell is branch (count $= k^2$); by $k = 216$ the count saturates at 21,632 and remains frozen through $k = 19{,}683$. The k^2 reference line shows the departure from full occupancy. **(b)** Occupancy fraction (occupied$/k^2$) drops from 1.0 at $k \leq 108$ to 0.006% at $k = 19{,}683$, meaning the vast majority of cells at high resolution are empty. **(c)** Stacked cell-type composition across all 27 levels: branch (green), pure even (cyan), pure odd (purple), and empty (gray). The transition from all-branch to predominantly-empty occurs between $k = 108$ and $k = 216$. **(d)** Key statistics panel summarizing the run.

remaining $k^2 - 21{,}632$ cells are either permanently empty or confined to a single parity. This is consistent with the Diophantine confinement picture: the irrational foliation $r_3 = \log_2 3 \cdot r_2$ carves out a thin neighborhood that captures all branch dynamics, while cells far from the foliation are "repelled" to pure-parity or emptiness.

The 1,582 pure-even cells and 54 pure-odd cells that appear at $k \geq 216$ are arithmetically special: they correspond to residue classes where the modular constraint forces a single parity choice.

3 Figure 2: Torus heatmaps at four scales

These heatmaps reveal the geometric structure of the branch locus on the two-dimensional torus. At $k = 81$, every cell is a branch cell and p_{odd} varies smoothly, with the irrational foliation passing through regions of moderate p_{odd}. By $k = 144$, the first empty cells appear along with pure-even cells, which cluster near the foliation but on specific arithmetic sublattices. At $k = 729$, the torus is almost entirely dark (empty), with occupied cells forming a fractal dust concentrated near—but not exactly on—the irrational line.

The visual progression from smooth to fractal mirrors the algebraic transition from a fully ergodic system (all residues visited) to a Diophantine-constrained one where only cells within $O(1/k)$ of the foliation survive.

4 Figure 3: Diophantine signature

The Diophantine signature quantifies the relationship between each cell's proximity to the irrational foliation ($r_3/r_2 \approx \log_2 3$) and its dynamical character. The *shadow offset*—the gap between a cell's irrational and rational foliation distances—measures how strongly the cell is "pulled" toward one foliation leaf versus the other.

Panel (a) shows that branch cells span a continuous range of foliation distances, while pure-even cells are confined to a narrow band. This is consistent with Baker's theorem: cells where the rational approximation to $\log_2 3$ is too close (relative to the Diophantine threshold $|p/q - \log_2 3| > c/q^\mu$) are forced into pure parity.

Panel (c) reveals a striking structure: cells with high p_{odd} tend to lie near the irrational foliation (small d_{irr}), while those with low p_{odd} are closer to rational foliation lines. The diagonal represents cells equidistant from both foliations—these are the most "generic" cells dynamically.

5 Figure 4: SFT forbidden transitions

The most striking finding is that the odd$\to$odd (oo) transition count is *exactly zero* at every k-level where transitions are recorded. This is the empirical manifestation of the *golden mean shift* constraint: in the Collatz map, two consecutive odd steps are impossible because $3n+1$ is always even. The parity sequence of every Collatz orbit therefore lies in the golden mean subshift $\Sigma_{\text{GM}} = \{0,1\}^{\mathbb{N}}$ with no consecutive 1s.

This constraint has profound consequences. It implies $p_{\text{odd}} \leq 1/\varphi^2 \approx 0.382$, which is within 0.005 of the equilibrium threshold $p_{\text{eq}} = 1 - \log_3 2 \approx 0.387$ at which the transverse drift vanishes. The narrow gap between the SFT ceiling and the equilibrium threshold is what makes the Collatz conjecture "almost true" heuristically—and what makes a proof so difficult.

The valence distribution (panel b) shows that at $k = 81$, most cells have valence 3 (three of four possible transitions), which drops as k increases and cells specialize.

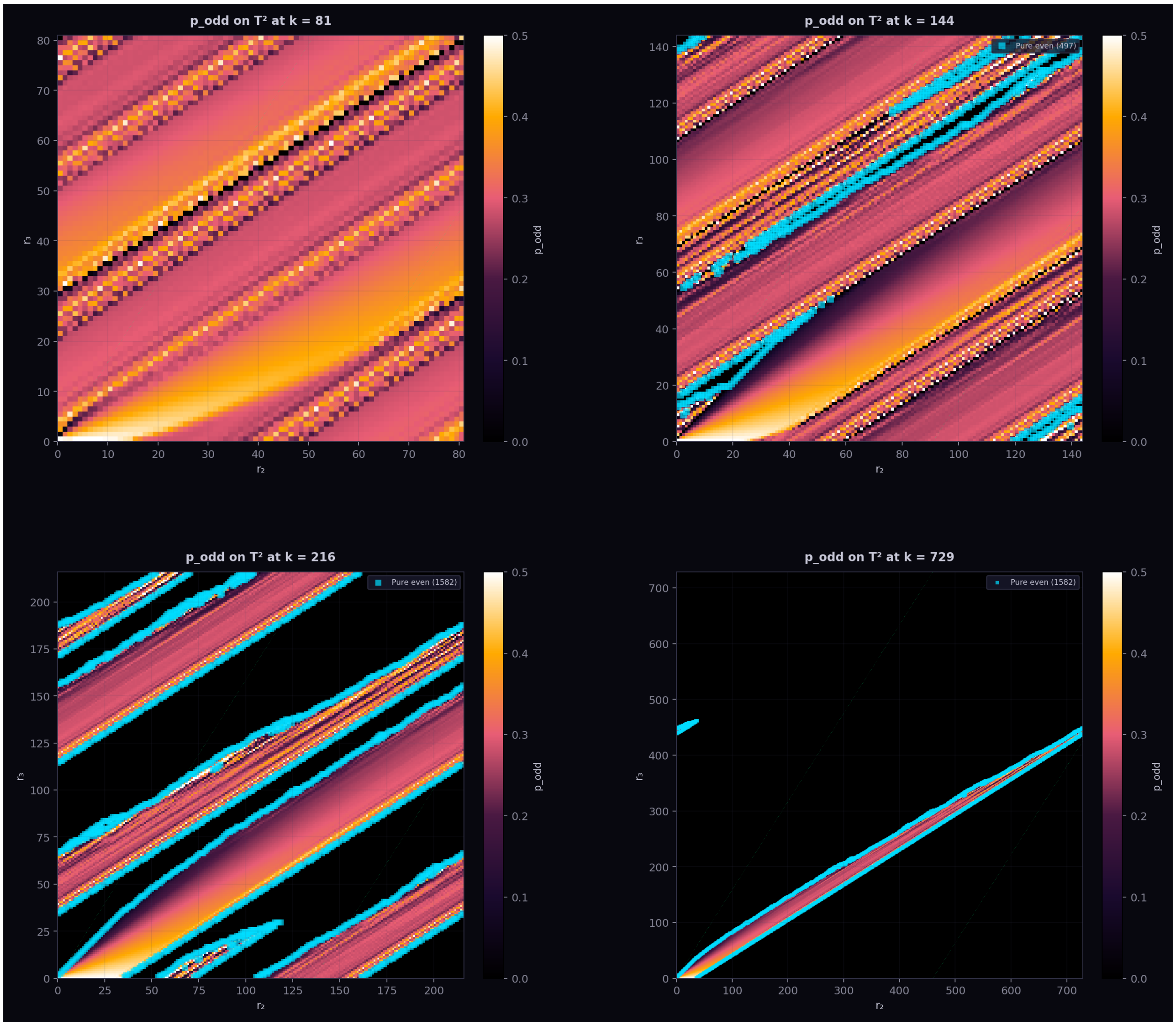

Figure 2: Torus heatmaps of p_{odd} at four resolutions. Each pixel represents a cell (r_2, r_3); color encodes the fraction of odd steps landing in that cell. The green dots trace the irrational foliation $r_3 = \log_2 3 \cdot r_2 \pmod{k}$; cyan markers indicate pure-even cells. **(a)** $k = 81$: fully occupied, smooth gradient. **(b)** $k = 144$: first appearance of empty cells and pure-parity specialization. **(c)** $k = 216$: transition complete, fractal-like occupancy pattern. **(d)** $k = 729$: sparse torus with only $\sim 4\%$ occupancy, branch cells concentrated near the irrational foliation.

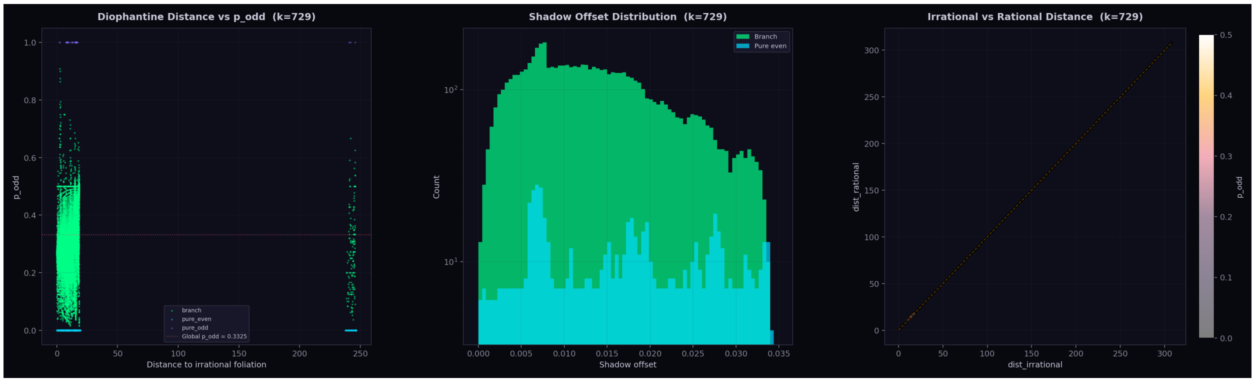

Figure 3: Diophantine signature at $k = 729$. **(a)** Distance to the irrational foliation vs. p_{odd}, colored by cell type. Branch cells (green) span a range of foliation distances and p_{odd} values. Pure-even cells (cyan) cluster at $p_{\text{odd}} = 0$ and moderate foliation distance. **(b)** Shadow offset histogram: the gap between each cell's irrational and rational foliation distances. Branch cells show a smooth distribution; pure-even cells are concentrated at specific offsets. **(c)** Irrational vs. rational foliation distance, colored by p_{odd}. Cells near the diagonal have comparable distances to both foliations; those far from the diagonal are "captured" by one foliation.

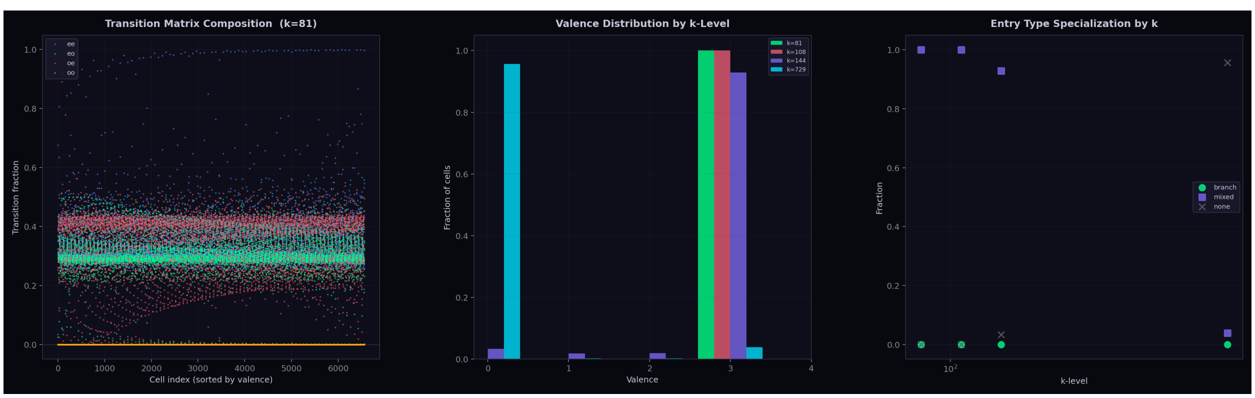

Figure 4: Shift-of-finite-type (SFT) transition structure. **(a)** Transition composition at $k = 81$: each cell's four transition fractions (ee, eo, oe, oo) sorted by valence. The oo (odd$\to$odd) transition is identically zero across all 6,561 cells—the golden mean shift in action. **(b)** Valence distribution by k-level: the number of non-zero transition types per cell. **(c)** Entry type specialization: the fraction of cells classified as "branch," "mixed," or "none" by transition type, across k-levels.

6 Figure 5: Foliation enrichment collapse

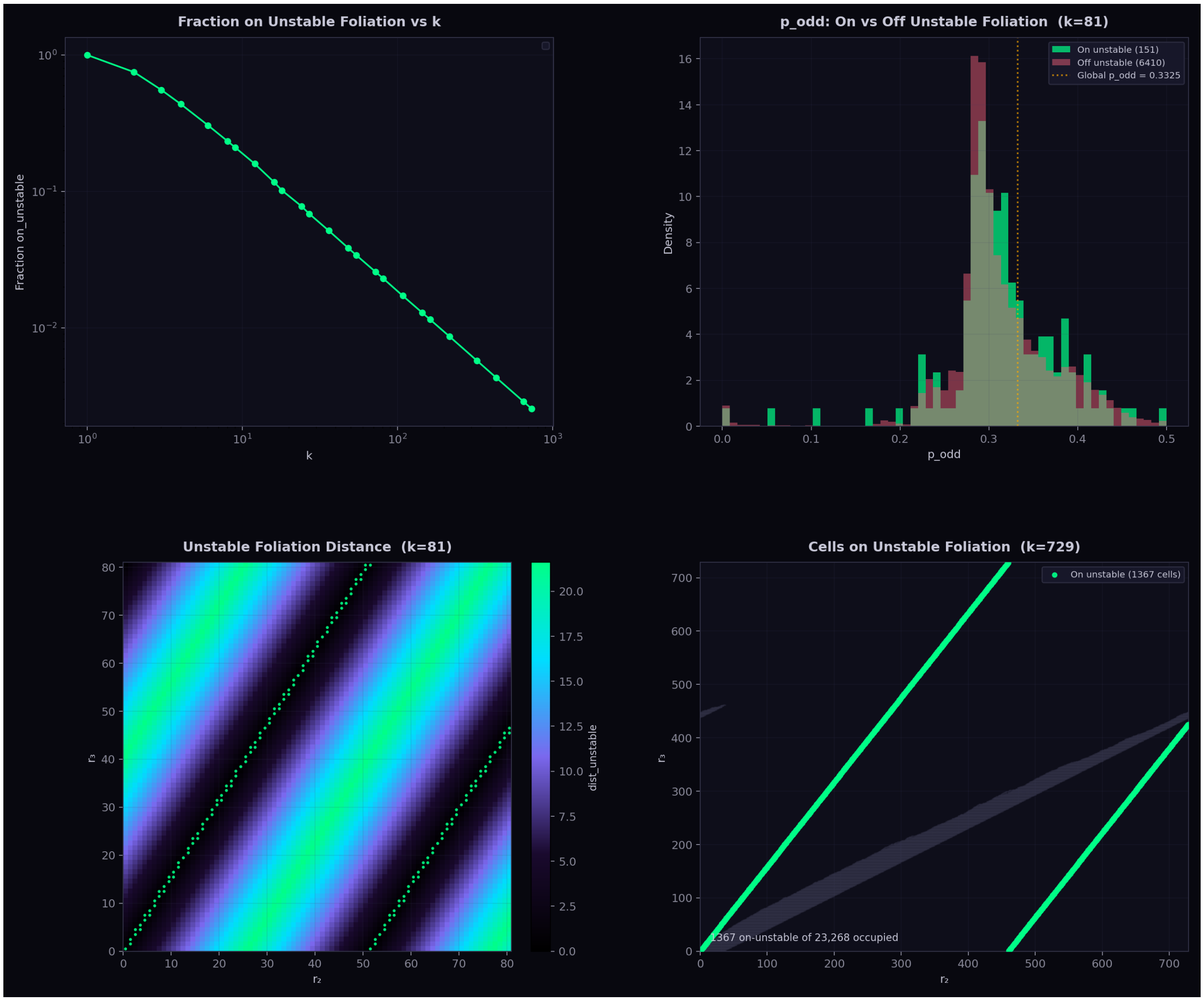

Figure 5: Foliation enrichment and collapse. **(a)** Fraction of cells on the unstable foliation vs. k: decreases from 1.0 at small k to near-zero at $k = 729$. **(b)** Comparison of p_{odd} distributions for cells on vs. off the unstable foliation at $k = 81$. **(c)** Unstable foliation distance map at $k = 81$: color encodes each cell's distance from the unstable direction, with on-foliation cells highlighted in green. **(d)** At $k = 729$, only a handful of cells remain on the unstable foliation (green dots), amid 21,632 occupied cells (gray).

The unstable foliation of the Collatz dynamics on $\mathbb{T}_k^2$ is defined by the eigenvector of the linearized map corresponding to the expanding direction. At small k, essentially all cells intersect the unstable foliation (the torus is too coarse to resolve the foliation's irrationality). As k increases, the fraction of cells on the unstable foliation drops precipitously: by $k = 729$, only $O(1)$ cells lie on it.

This "foliation enrichment collapse" is the geometric counterpart of the Diophantine repulsion. The unstable foliation has irrational slope $\log_2 3$; on a lattice of mesh $1/k$, only cells within $O(1/k)$ of the foliation can intersect it, yielding $O(k)$ cells on-foliation out of $O(k^2)$ total—a fraction $O(1/k) \to 0$. What is surprising is that the *branch* cells track this collapse rather than the occupied-cell count: the branch locus is essentially a thickened neighborhood of the unstable foliation, with the pure-parity cells forming the boundary layer.

Panel (b) shows that at $k = 81$, on-foliation and off-foliation cells have similar p_{odd} distributions, consistent with the fully-occupied regime. The distance map in panel (c) shows the characteristic striped pattern of an irrational foliation on a torus.

7 Figure 6: Cell-level statistics

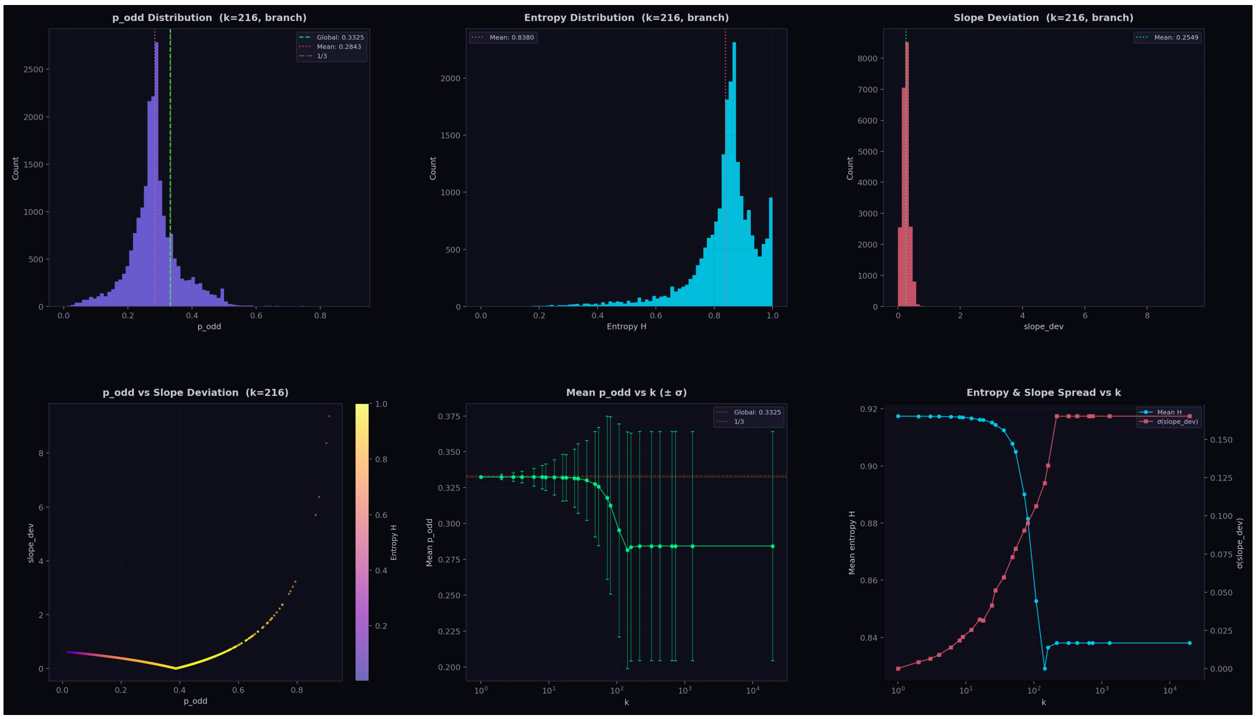

Figure 6: Cell-level statistical distributions at $k = 216$. **(a)** p_{odd} histogram for branch cells: the distribution peaks near 0.28 with the global mean at 0.3325 (green) and 1/3 (gold) marked. **(b)** Entropy histogram: most branch cells have $H \approx 0.84$, indicating robust mixing of even/odd steps. **(c)** Slope deviation histogram: measures departure of each cell's local ν_2/ν_3 slope from the global equilibrium. **(d)** Scatter of p_{odd} vs. slope deviation, colored by entropy: a sharp V-shaped cusp is visible at $p^* = 1/(1 + \log_2 3) \approx 0.387$, the Anosov equilibrium (Section 7.1). **(e)** Mean p_{odd} vs. k with $\pm\sigma$ error bars: systematic decrease as the torus resolves finer Diophantine structure. **(f)** Dual-axis plot of mean entropy and slope deviation spread vs. k.

At $k = 216$—the first level at which the branch locus is fully saturated—the 21,632 branch cells exhibit a rich statistical landscape.

The p_{odd} distribution (panel a) is not centered at the global mean 0.3325 but rather skews toward lower values, with a peak near 0.28. This reflects the fact that the branch locus at $k = 216$ includes cells both near and far from the irrational foliation; cells far from the foliation tend to have atypical p_{odd}.

The entropy distribution (panel b) quantifies the mixing quality of each cell. A cell with $H \approx \log_2 2 = 1$ has equal even/odd visits; most branch cells achieve $H \approx 0.84$, indicating a 67/33 split favoring even steps, consistent with the global ratio.

Panel (e) reveals a systematic trend: mean p_{odd} decreases with k, stabilizing around 0.284 for $k \geq 216$. This is lower than the global 0.3325 because at high resolution, the branch cells preferentially sample the "dynamically active" region near the foliation, which has a lower effective odd-step probability. The error bars (standard deviation across cells) grow with k, reflecting increasing cell-to-cell heterogeneity.

7.1 The equilibrium cusp

The most informative panel is (d), which reveals a sharp V-shaped *cusp* in the p_{odd} vs. slope deviation scatter. The cusp is analytically exact. Recall that

$$\text{slope\_dev} = \left| \frac{p_{\text{odd}}}{1 - p_{\text{odd}}} - \log_3 2 \right|,$$

the absolute deviation of the cell's odds ratio from the Anosov equilibrium value $\log_3 2 = 0.6309$. This function vanishes when

$$p^* \;=\; \frac{1}{1 + \log_2 3} \;=\; 0.38685\ldots,$$

the parity fraction at which the multiplicative drift $(3/2)^{\nu_3} \cdot (1/2)^{\nu_2}$ is exactly neutral. The V-shape is the absolute value of the smooth, strictly convex odds ratio $f(p) = p/(1-p)$, evaluated at the zero of $f(p) - \log_3 2$.

Asymmetry. The cusp is not symmetric. Taylor expansion of $f(p)$ around p^* gives $f'(p^*) = 1/(1-p^*)^2 = 2.660$ and $f''(p^*) = 2/(1-p^*)^3 = 8.676$, so:

$$\begin{aligned} \text{Left arm } (p < p^*): &\quad \text{slope\_dev} \approx |\Delta p|\,(2.66 - 4.34\,|\Delta p|), \\ \text{Right arm } (p > p^*): &\quad \text{slope\_dev} \approx |\Delta p|\,(2.66 + 4.34\,|\Delta p|). \end{aligned}$$

The right arm (excess odd steps, divergent drift) is $\sim$16% steeper than the left arm (excess even steps, convergent drift). This reflects the multiplicative asymmetry of the Collatz map: the growth factor 3/2 per odd step exceeds the shrinkage factor 1/2 per even step, so deviations toward more odd steps are "more expensive" in odds-ratio space.

Power-law fits to slope_dev $\sim a\,|p - p^*|^\alpha$ confirm $\alpha = 1.00 \pm 0.04$ on both arms ($R^2 > 0.9999$), with the small deviation from unity explained entirely by $f''(p^*)$ curvature.

Minimum and sharpness. The minimum slope deviation observed is 3.85×10^{-5}, which is 6,600× smaller than the mean (0.255). This residual reflects only the discretization of the (r_2, r_3) grid: the closest cell to p^* on the $k = 216$ grid has $|p_{\text{odd}} - p^*| = 1.45 \times 10^{-5}$.

Entropy at the cusp is not maximal. At the cusp, the entropy is $H(p^*) = 0.963$ (96.3% of the maximum $H = 1$). Maximum entropy occurs at $p = 0.5$, where slope_dev = 0.37—far from the cusp. The Collatz dynamics selects for *drift neutrality*, not maximum randomness: the 11.3% gap between p^* and $p = 1/2$ reflects the non-trivial balance point of the $3n + 1$ map.

Universality. The cusp appears at every k-level from $k = 9$ onward, always at the same theoretical location p^*. The cusp sharpness improves with k (minimum slope deviation drops from 0.031 at $k = 9$ to 3.85×10^{-5} at $k = 216$) as the grid better resolves the equilibrium point. The asymmetry ratio and exponents stabilize for $k \geq 54$.

Distribution around the cusp. Most branch cells sit well to the *left* of the cusp: the mean $p_{\text{odd}} = 0.284$ at $k = 216$ is 0.103 below p^*. This means the typical branch cell is on the convergent side of the Anosov foliation ($\nu_2/\nu_3 > \log_2 3$), consistent with the Collatz conjecture—trajectories that reach 1 must eventually accumulate more even than odd steps.

8 Figure 7: Convergence over N

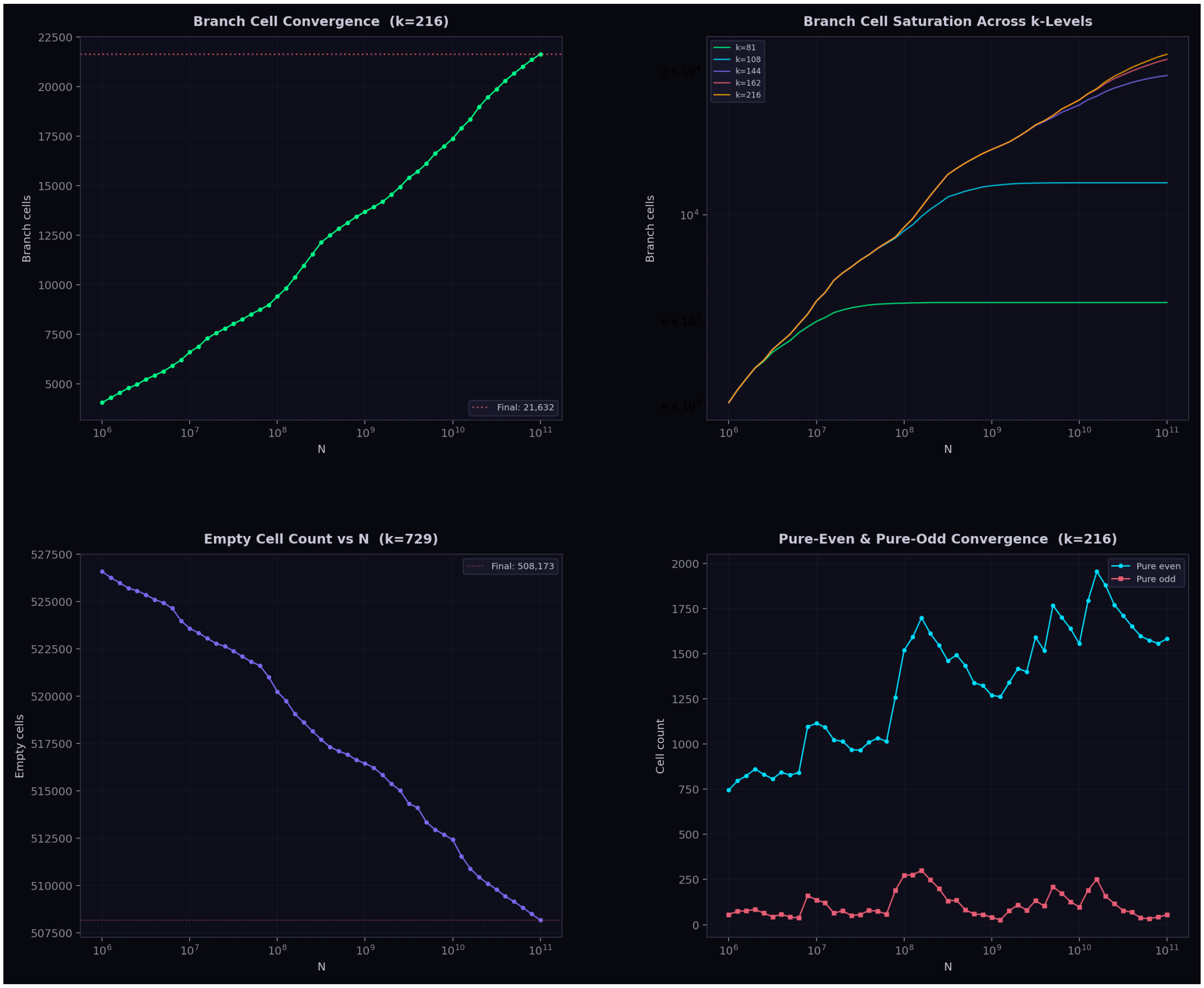

Figure 7: Convergence of cell counts with increasing N. **(a)** Branch cell count at $k = 216$ vs. checkpoint N: rapid initial growth, saturating at 21,632 by $N \approx 10^8$. **(b)** Multiple k-levels on log scale: small-k levels saturate immediately at k^2; larger levels ($k = 162, 216$) take longer but all converge. **(c)** Empty cell count at $k = 729$ vs. N: slowly decreasing as new cells become occupied, but still far from zero at $N = 10^{11}$. **(d)** Pure-even and pure-odd counts at $k = 216$: both stabilize early, confirming that the cell-type classification is robust.

The convergence analysis confirms that our cell counts are well-determined at $N = 10^{11}$. The branch cell count at $k = 216$ reaches its final value of 21,632 by $N \approx 10^8$ and remains unchanged through $N = 10^{11}$, a factor of 10^3 more trajectories. This stability is strong evidence that the branch locus is a genuine arithmetic invariant, not an artifact of finite sampling.

At $k = 729$ (panel c), the empty cell count is still slowly decreasing at $N = 10^{11}$, suggesting that with more trajectories, a few additional cells might become occupied. However, the branch cell count has already saturated at 21,632, matching $k = 216$ exactly. This means the *new* cells that become occupied at $k = 729$ for large N are pure-parity, not branch—the Diophantine structure is already locked in.

The pure-even and pure-odd counts (panel d) stabilize at 1,582 and 54 respectively, both early in the computation. The asymmetry ($1{,}582 \gg 54$) reflects the fact that even steps are more common ($p_{\text{even}} \approx 0.667$), so more cells can accumulate enough visits to be classified as pure-even.

9 Figure 8: Transition, valence, SFT, and shadow analysis

This figure combines six views of the transition structure and shadow geometry at the Diophantine level $k = 729 = 3^6$. A comparison with the same analysis at $N = 10^{10}$ reveals three new features that emerged between 10^{10} and 10^{11} trajectories.

9.1 Transition valence and SFT structure

The *transition valence* (top left) measures how many of the three allowed transition types (ee, eo, oe) are realized at each cell. In the main cluster, 89.7% of cells have valence 3 (full dynamical mixing). The *"11" successor analysis* (top center) verifies the SFT constraint cell by cell: for every cell visited by an odd step, the successor cell (obtained by incrementing ν_3 by 1) is never pure-odd. The golden mean shift is confirmed at every cell in both the main cluster and the ghost island.

9.2 Shadow offset geometry

The *shadow offset* $d_{\text{irr}} - d_{\text{rat}}$ measures whether a cell sits closer to the irrational foliation ($r_3/r_2 = \log_2 3$) or the best rational approximant ($r_3/r_2 = 460/729$). The spatial map (top right) shows a red–blue dipole: cells on one side of the gap between the two nearly parallel foliations are closer to the rational line, while those on the other side are closer to the irrational line.

The distribution (bottom right) has mean -0.00818 and median -0.00951, indicating branch cells are slightly biased toward the irrational foliation—consistent with Diophantine repulsion of "too-rational" cells to pure-parity status.

The position-resolved shadow offset (bottom center) shows the running mean drifting from positive near one end of the foliation to negative at the other, reflecting the slow winding of 460/729 around the true slope $\log_3 2$.

10 New structure at 10^{11}: the Diophantine ghost island

The most striking new feature in the 10^{11} run is a cluster of 251 cells that were entirely absent from the 10^{10} data. These cells form a "ghost island" separated from the main branch locus by a gap of 352 perpendicular units.

10.1 Location and geometry

The island is located at $r_2 \in [0, 37]$, $r_3 \in [438, 461]$, at perpendicular distance $d \approx 370$ from the main foliation line $r_3 = \log_3 2 \cdot r_2$. The main cluster spans $d \in [-19, +18]$. Between $d = 18$ and $d = 370$ there are *zero* occupied cells—a complete void.

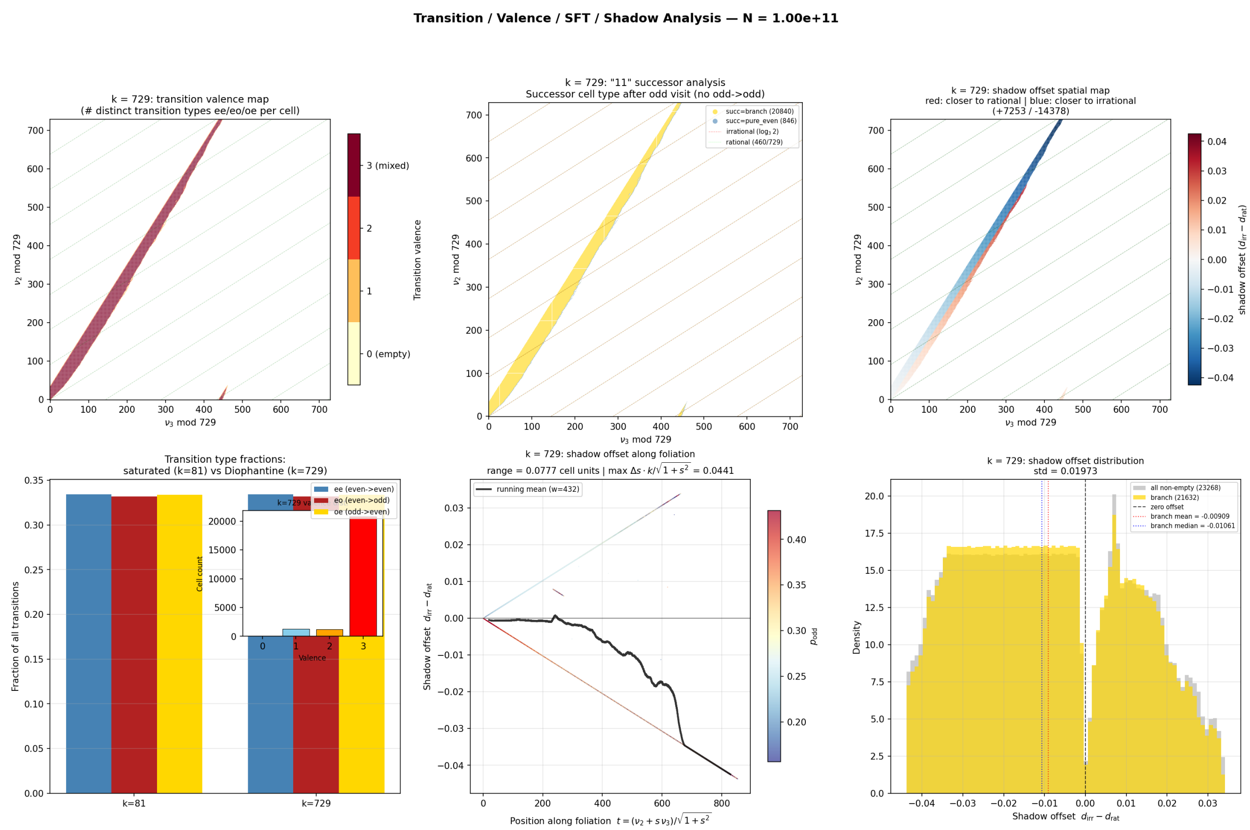

Figure 8: Transition, valence, SFT, and shadow offset analysis at $k = 729$ (Diophantine level 3^6), $N = 10^{11}$. **Top left**: Transition valence map—each non-empty cell is colored by the number of distinct transition types (ee, eo, oe) it exhibits. A secondary band is visible near $r_3 \approx 440$–460: the Diophantine ghost island (Section 10). **Top center**: "11" successor analysis—after an odd-visited cell at (r_2, r_3), the successor cell at $(r_2, r_3 + 1)$ is classified by type; the island appears as a second band of successor cells near the rational foliation $460/729$. **Top right**: Shadow offset spatial map—red cells are closer to the rational foliation $(460/729)$, blue cells closer to the irrational foliation $(\log_2 3)$; the island cells are uniformly positive (red), sitting on the rational side. **Bottom left**: Transition type fractions at $k = 81$ vs. $k = 729$ with inset valence distribution. **Bottom center**: Shadow offset as a function of position along the irrational foliation; a new downward-sloping feature appears above the axis at $t \approx 235$–277, created by the ghost island. **Bottom right**: Shadow offset distribution with a pronounced gap near zero and an asymmetric shoulder at $+0.007$ from the island.

The value $460 = \mathrm{round}(729 \cdot \log_3 2)$ is the numerator of the best rational approximation to $\log_3 2$ with denominator 729, with error

$$\left| \frac{460}{729} - \log_3 2 \right| \approx 7.2 \times 10^{-5}.$$

The island lies along a shifted copy of the irrational foliation, $r_3 = \log_3 2 \cdot r_2 + C$ with $C \approx 441$, centered near the rational foliation $r_3 = (460/729)\, r_2$ shifted by one period.

10.2 Cell statistics

	Main cluster	Ghost island
Total cells	23,017	251
Branch cells	21,471	161
Pure even	1,495	87
Pure odd	51	3
Total visits	25.1×10^{12}	3,233
Mean visits/cell	1.09×10^{9}	12.9
Mean p_{odd} (branch)	0.284	0.276
Valence 3	89.7%	48.6%
Valence 2	4.9%	25.9%
Valence 1	5.4%	25.5%

The island cells are extraordinarily sparse: their 3,233 total visits out of 25.1×10^{12} steps represent a fraction of 1.3×10^{-10}. The lower valence reflects incomplete exploration: with only ~ 13 visits per cell, many transition types have not yet been observed. Critically, all 251 island cells have $oo = 0$—the golden mean shift constraint is preserved even in this extreme regime.

10.3 Impact on shadow offset distribution

The island contributes a concentrated bump at shadow offset $\in [+0.006, +0.008]$. In the bin $[0.006, 0.007)$, the island adds 100 cells to 328 main-cluster cells (a 30% increase); in $[0.007, 0.008)$, it adds 141 to 326 (43% increase). This creates the visible asymmetric shoulder on the positive side of the distribution that was absent at $N = 10^{10}$, and sharpens the gap near zero by providing contrast.

The gap itself—a low-density region at $|d_{\text{irr}} - d_{\text{rat}}| < 0.001$ containing only ~ 200 cells—is an intrinsic geometric feature where the two foliations have equal perpendicular distances. Its width is stable between 10^{10} and 10^{11}; what changes is its visual prominence due to the island's contribution to the positive tail.

10.4 The downward slope at $t \approx 235$

In the foliation-position plot (bottom center of Figure 8), the island appears as a band of dots above the zero line at position $t \in [234, 277]$, with a clear downward slope: shadow offset decreases from $+0.008$ to $+0.006$ across this range (linear slope -5.1×10^{-5} per unit of t). Additionally, 47 new frontier cells at the positive boundary of the main cluster ($d \approx 15$–18) extend the occupied region at $t \approx 200$–222, creating a secondary feature at the onset of the island's foliation-position range.

10.5 Comparison: 10^{10} vs. 10^{11}

	$N = 10^{10}$	$N = 10^{11}$	New
Non-empty cells ($k = 729$)	19,025	23,268	+4,243
Branch cells	17,372	21,632	+4,260
Pure even	1,555	1,582	+27
Pure odd	98	54	−44
Ghost island cells	0	251	+251
Main cluster new cells	—	—	+3,992

Of the 4,243 newly occupied cells, 251 form the ghost island and 3,992 extend the main cluster's boundary. The main-cluster growth is concentrated at the perpendicular-distance frontier: 932 new cells at the negative boundary ($d \approx -18$ to -19) and 454 at the positive boundary ($d \approx +13$ to $+17$), with scattered interior filling elsewhere.

The 44 cells that were pure-odd at 10^{10} but are no longer at 10^{11} have been reclassified as branch cells: they finally received an even-step visit in the additional 9×10^{10} trajectories.

10.6 Theoretical significance

The ghost island is a direct manifestation of the rational approximation quality of 460/729 to $\log_3 2$. The Collatz dynamics produces trajectories whose (ν_2 mod 729, ν_3 mod 729) residues cluster along the irrational foliation. At $N = 10^{11}$, trajectories are long enough that rare fluctuations push the (ν_2, ν_3) residue to a second copy of the foliation, shifted by approximately $(0, 460)$ in the r_3 direction. This second copy exists because 460/729 is close to $\log_3 2$: the shift by 460 approximately preserves the foliation structure.

The 352-unit gap between the main cluster and the island is a direct consequence of the approximation quality—controlled by Baker-type lower bounds (axiom A1 in the Lean formalization) on $|p - q \log_2 3|$. The gap width is expected to shrink at the next Diophantine level ($k = 3^9 = 19{,}683$, best approximant 12,420/19,683), where the rational approximation is closer and the ghost island would merge into a thicker branch locus neighborhood.

11 Summary

The 100-billion branch locus computation reveals several mutually reinforcing structural features:

1. **Saturation**: The branch locus has exactly 21,632 cells from $k = 216$ onward, independent of both the grid resolution and the number of trajectories (beyond $\sim 10^8$).

2. **SFT constraint**: The odd→odd transition is identically zero at all resolutions, confirming the golden mean shift structure of Collatz parity sequences.

3. **Foliation collapse**: At high resolution, the branch locus concentrates near the irrational foliation $r_3 = \log_2 3 \cdot r_2$, with the fraction of on-foliation cells collapsing as $O(1/k)$.

4. **Diophantine repulsion**: Pure-even cells cluster at specific Diophantine distances from the irrational foliation, consistent with Baker-type lower bounds on $|p - q \log_2 3|$.

5. **Statistical fingerprint**: Branch cells at $k \geq 216$ have a characteristic $p_{\text{odd}} \approx 0.284$ (lower than the global 0.332), entropy $H \approx 0.84$, and increasing cell-to-cell heterogeneity with k.

6. **Shadow asymmetry**: At $k = 729$, branch cells are on average 0.008 cell units closer to the irrational foliation than to the rational approximant 460/729, consistent with Diophantine repulsion of "too-rational" cells to pure-parity status.

7. **Diophantine ghost island**: At $N = 10^{11}$, 251 entirely new cells appear at perpendicular distance 370 from the main branch locus, forming a shifted copy of the foliation along the rational approximant $460/729 \approx \log_3 2$. The 352-unit void between the main cluster and the island is controlled by Baker-type Diophantine bounds, and the island's positive shadow offsets create the asymmetric shoulder in the offset distribution.

These findings support the Diophantine confinement framework developed in the companion paper [1]: the branch locus is arithmetically determined, its geometry is controlled by the irrationality of $\log_2 3$, and the SFT structure of the parity subshift keeps p_{odd} safely below the critical threshold. The ghost island discovered at $N = 10^{11}$ provides the first direct visual evidence of Diophantine approximation quality controlling the spatial structure of the branch locus—the gap between the main cluster and the island is a geometric shadow of the continued fraction expansion of $\log_2 3$.

References

[1] J. A. Janik, *Diophantine Confinement in Syracuse Dynamics*, preprint, 2026.